

Physical Chemistry (I)

Lecture 11



5. Helmholtz and Gibbs Energies

Chapter Overview

- Helmholtz and Gibbs energies
 - Definitions
 - Meanings
- Thermodynamics potentials
 - Differential forms
 - Properties of exact differentials
- Thermochemistry of Gibbs energy

Clausius Inequality

$$dS \geq \frac{\delta q}{T}$$

$$dS - \frac{\delta q}{T} \geq 0$$

1. Constant V :

$$dS - \frac{dU}{T} \geq 0$$

2. Constant p :

$$dS - \frac{dH}{T} \geq 0$$

Spontaneity

1. Constant V :

$$dS - \frac{dU}{T} \geq 0$$

$$\underline{d(U - TS)} \leq 0 \text{ (at constant } T\text{)}$$

Helmholtz energy A

2. Constant p :

$$dS - \frac{dH}{T} \geq 0$$

$$\underline{d(H - TS)} \leq 0 \text{ (at constant } T\text{)}$$

Gibbs energy G

Helmholtz and Gibbs Energies

$$\text{Helmholtz energy } A = U - TS$$

$$\text{Gibbs energy } G = H - TS$$

Criteria of spontaneity:

$$dA_{T,V} \leq 0$$

(At constant T and V , $dA \propto -dS_{\text{total}}$)

$$dG_{T,p} \leq 0$$

(At constant T and p , $dG \propto -dS_{\text{total}}$)

So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work

A and Maximum Work

$$TdS \geq \delta q \text{ (Clausius inequality)}$$

$$dU = \delta q + \delta w \text{ (first law)}$$

Hence,

$$TdS \geq dU - \delta w$$

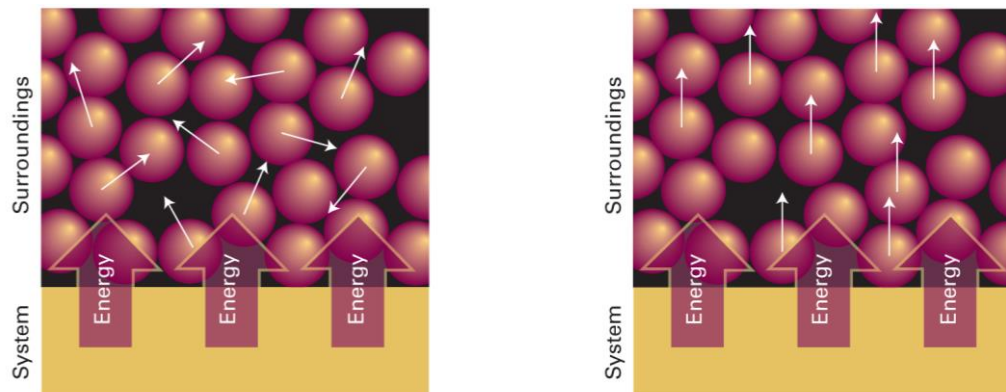
$$\delta w \geq dU - TdS = dA_T$$

As $\delta w < 0$ for the work by the system,

$$dA_T = \delta w_{\max}$$

$$\Delta A_T = w_{\max}$$

Meaning of Helmholtz Energy



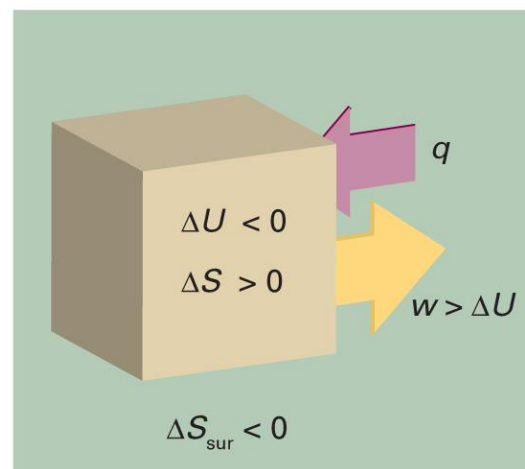
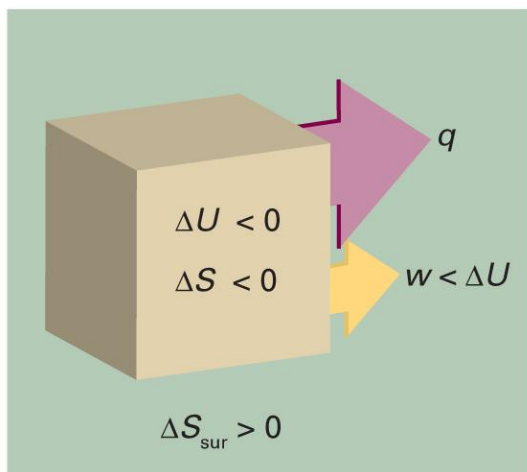
$$A = U - TS$$

At temperature T ,

TS = energy stored as thermal motion

$U - TS$ = maximum work obtainable from the system

ΔS and Maximum Work



***G* and Maximum Non-Exp Work**

$$H = U + pV$$

$$dH = dU + d(pV)$$

$$dH = \delta q + \delta w + d(pV) \text{ (first law)}$$

Recall $G = H - TS$. At constant T ,

$$dG_T = dH - TdS = \delta q + \delta w + d(pV) - TdS$$

For a reversible change, $\delta q_{\text{rev}} = TdS$

$$dG_T = \delta w_{\text{rev}} + d(pV)$$

G and Maximum Non-Exp Work

$$dG_T = \delta w_{\text{rev}} + d(pV)$$

At constant p ,

$$dG_{T,p} = \delta w_{\text{rev}} + pdV$$

On the other hand,

$$\delta w_{\text{rev}} = -pdV + \delta w_{\text{add,rev}}$$

Thus,

$$dG_{T,p} = \delta w_{\text{add,rev}} = \delta w_{\text{add,max}}$$

$$\Delta G_{T,p} = w_{\text{add,rev}} = w_{\text{add,max}}$$

Example 3D.1

When 1.000 mol $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) is oxidized completely to carbon dioxide and water at 25°C according to the equation $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$, calorimetric measurements give $\Delta_r U = -2808 \text{ kJ mol}^{-1}$ and $\Delta_r S = +182.4 \text{ J K}^{-1} \text{ mol}^{-1}$ at 25°C and 1 bar. How much of this change in internal energy can be extracted as (a) heat at constant pressure, (b) work?

a) Heat extractable at constant pressure = $-\Delta H$

$$\Delta_r H = \Delta_r U + \Delta_r(pV) \approx \Delta_r U + RT\Delta_r \nu_g$$

Since $\Delta_r \nu_g = 6 - 6 = 0$, $\Delta_r H \approx \Delta_r U = -2808 \text{ kJ mol}^{-1}$

Thus, the maximum extractable heat is **2808 kJ mol^{-1}** .

Example 3D.1

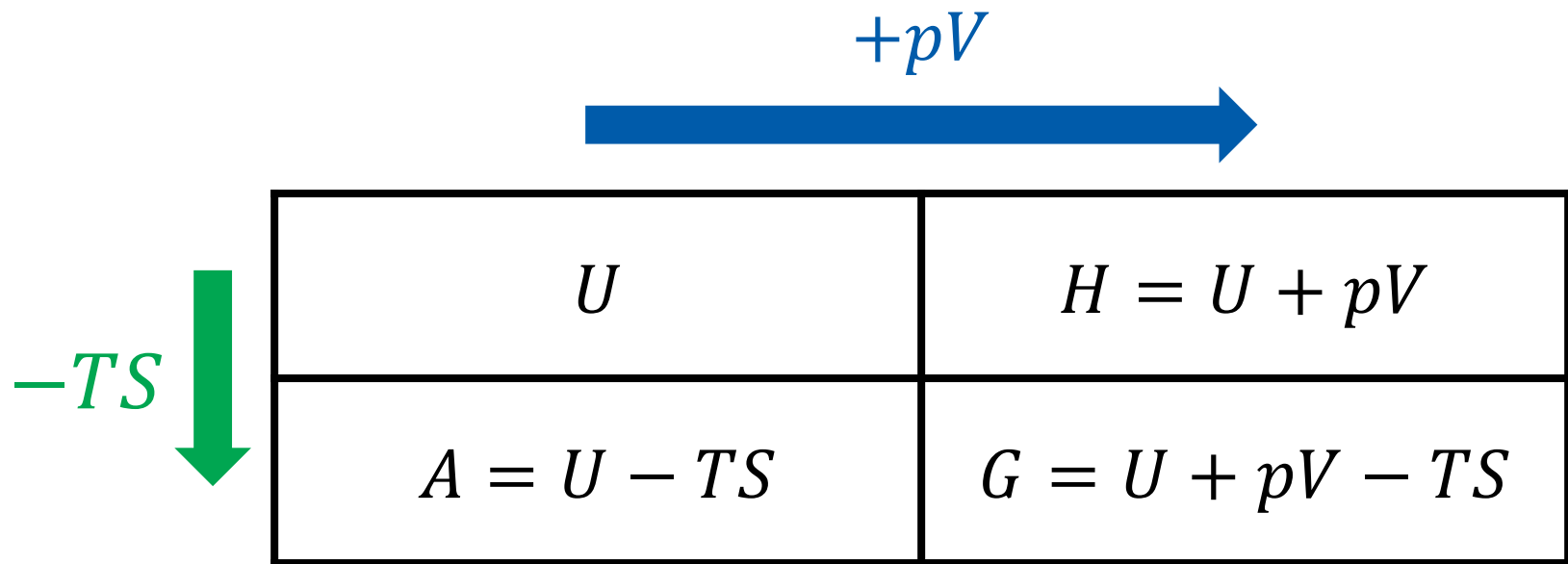
When 1.000 mol $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) is oxidized completely to carbon dioxide and water at 25°C according to the equation $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$, calorimetric measurements give $\Delta_r U = -2808 \text{ kJ mol}^{-1}$ and $\Delta_r S = +182.4 \text{ J K}^{-1} \text{ mol}^{-1}$ at 25°C and 1 bar. How much of this change in internal energy can be extracted as (a) heat at constant pressure, (b) work?

b) Work extractable at constant pressure = $-\Delta A$

$$\begin{aligned}\Delta_r A &= \Delta_r U - T\Delta_r S = (-2808 \text{ kJ mol}^{-1}) - (298 \text{ K}) \times (182.4 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &= -2862 \text{ kJ mol}^{-1}\end{aligned}$$

Thus, the maximum extractable work is **2862 kJ mol^{-1}** .

Thermodynamic Potentials



Differential Forms

$$dU = TdS - pdV \text{ (fundamental equation)}$$

- Enthalpy $H = U + pV$

$$dH = dU + pdV + Vdp = (TdS - \cancel{pdV}) + \cancel{pdV} + Vdp$$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS$

$$dA = dU - TdS - SdT = (\cancel{TdS} - pdV) - \cancel{TdS} - SdT$$

$$dA = -SdT - pdV$$

Differential Forms

$$dU = TdS - pdV \text{ (fundamental equation)}$$

- Gibbs energy $G = U + pV - TS$

$$dG = dU + pdV + Vdp - TdS - SdT$$

$$dG = (\cancel{TdS} - \cancel{pdV}) + \cancel{pdV} + Vdp - \cancel{TdS} - SdT$$

$$dG = -SdT + Vdp$$

Natural Independent Variables

- Internal energy $U = U(S, V)$

$$dU = TdS - pdV$$

- Enthalpy $H = U + pV = H(S, p)$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$dA = -SdT - pdV$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$dG = -SdT + Vdp$$

Exact Differentials

The thermodynamic potentials are state functions

For $H = H(S, p)$,

$$dH = \left(\frac{\partial H}{\partial S}\right)_p dS + \left(\frac{\partial H}{\partial p}\right)_S dp$$

Comparing this to $dH = TdS + Vdp$,

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \quad \left(\frac{\partial H}{\partial p}\right)_S = V$$

Exact Differentials

Similarly, using $A = A(T, V)$, one can show

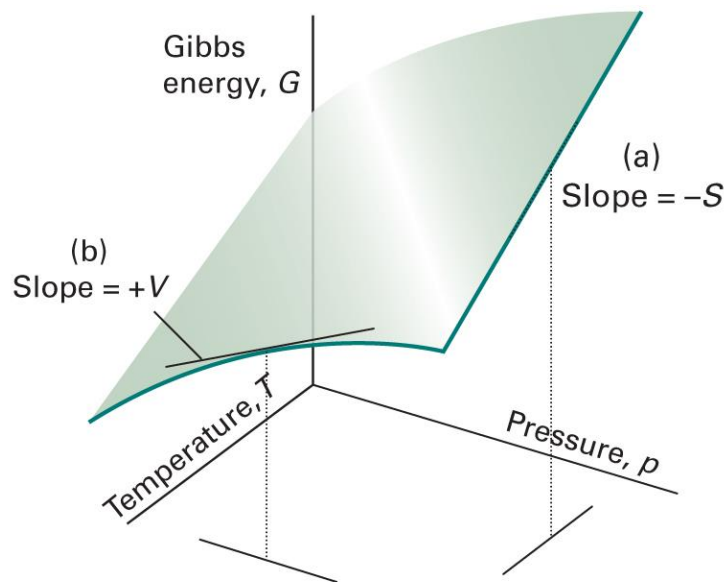
$$\left(\frac{\partial A}{\partial T}\right)_V = -S, \quad \left(\frac{\partial A}{\partial V}\right)_T = -p$$

Using $G = G(T, p)$, one can show

$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \quad \left(\frac{\partial G}{\partial p}\right)_T = V$$

Example: Gibbs Energy

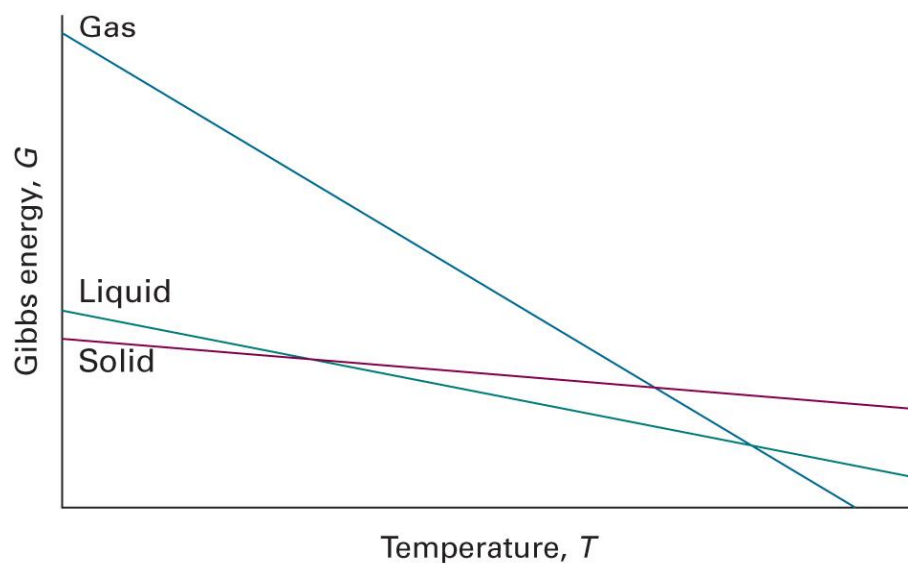
$$\left(\frac{\partial G}{\partial T}\right)_p = -S, \quad \left(\frac{\partial G}{\partial p}\right)_T = V$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 3E.1)

Physical Interpretation

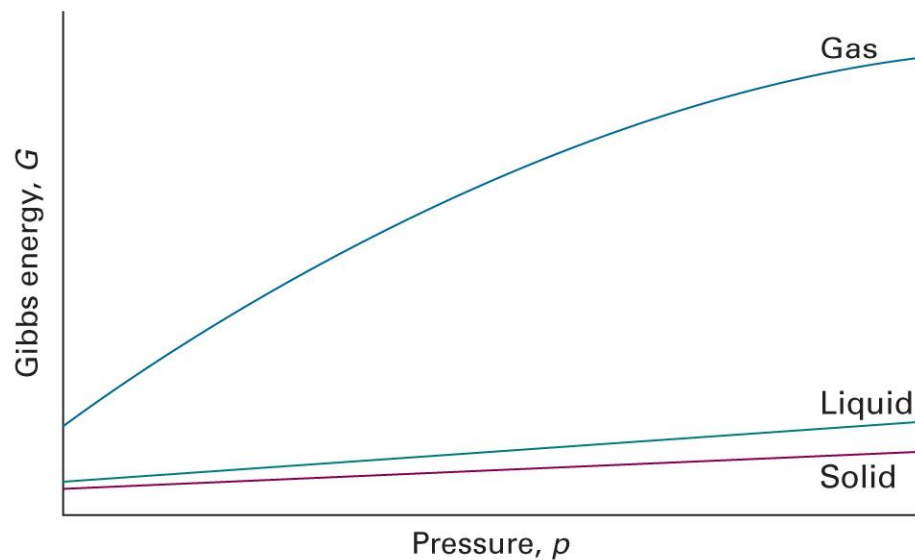
$$\left(\frac{\partial G}{\partial T}\right)_p = -S < 0$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 3E.2)

Physical Interpretation

$$\left(\frac{\partial G}{\partial p}\right)_T = V > 0$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 3E.3)

G and T

$$\left(\frac{\partial G}{\partial T}\right)_p = -S$$

Since $G = H - TS$,

$$\left(\frac{\partial G}{\partial T}\right)_p = \frac{G - H}{T}$$

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \left[\frac{\partial}{\partial T}\left(G \times \frac{1}{T}\right)\right]_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p + G \left[\frac{\partial}{\partial T}\left(\frac{1}{T}\right)\right]_p$$

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p - \frac{G}{T^2}$$

Since $(\partial G/\partial T)_p = (G - H)/T$,

$$\left(\frac{\partial(G/T)}{\partial T}\right)_p = \frac{G - H}{T^2} - \frac{G}{T^2} = -\frac{H}{T^2}$$

Application of G-H Equation

$$\left(\frac{\partial(G/T)}{\partial T} \right)_p = -\frac{H}{T^2}$$

At constant T ,

$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_p = -\frac{\Delta H}{T^2}$$

Property 1 of Exact Differential

If f is a function of x and y , the differential df is

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy$$

when df is an exact differential.

Then, we have

$$\left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)_y \right)_x = \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)_x \right)_y$$

(Schwartz's theorem)

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $dU = TdS - pdV$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $dH = TdS + Vdp$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $dA = -SdT - pdV$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $dG = -SdT + Vdp$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Example 3E.1

Use the Maxwell relations to show that the entropy of a perfect gas is linearly dependent on $\ln V$, that is, $S = a + b \ln V$.

From the Maxwell relations,

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T} \left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

Hence,

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

Example 3E.1

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V}$$

At constant temperature,

$$dS = \frac{nR}{V} dV$$

$$\int dS = \int \frac{nR}{V} dV$$

Therefore,

$$S = nR \ln V + C$$

Property 2 of Exact Differential

If f is a function of x and y , the differential df is

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy$$

when df is an exact differential.

Differentiating with respect to x at constant z gives

$$\left(\frac{\partial f}{\partial x} \right)_z = \left(\frac{\partial f}{\partial x} \right)_y + \left(\frac{\partial f}{\partial y} \right)_x \left(\frac{\partial y}{\partial x} \right)_z$$

Internal Pressure

Internal pressure $\pi_T = (\partial U / \partial V)_T$

From property 2,

$$\left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{\partial U}{\partial V}\right)_S + \left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T$$

Since $dU = TdS - pdV$,

$$\pi_T = -p + T \left(\frac{\partial S}{\partial V}\right)_T$$

Thermodynamic Eq. of State

$$\pi_T = T \left(\frac{\partial S}{\partial V} \right)_T - p$$

From the Maxwell relation $(\partial S / \partial V)_T = (\partial p / \partial T)_V$,

$$\pi_T = T \left(\frac{\partial p}{\partial T} \right)_V - p$$

or,

$$p = T \left(\frac{\partial p}{\partial T} \right)_V - \pi_T$$

(thermodynamic equation of state)

Example 3E.2

Show thermodynamically that $\pi_T = 0$ for a perfect gas, and compute its value for a van der Waals gas.

For a perfect gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T} \left(\frac{nRT}{V}\right)\right]_V = \frac{nR}{V}$$

From the thermodynamic equation of state,

$$\pi_T = T \left(\frac{\partial p}{\partial T}\right)_V - p = T \left(\frac{nR}{V}\right) - p = \frac{nRT}{V} - p = p - p = 0$$

Example 3E.2

For a van der Waals gas,

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left[\frac{\partial}{\partial T} \left(\frac{nRT}{V - nb} - a \frac{n^2}{V^2}\right)\right]_V = \frac{nR}{V - nb}$$

From the thermodynamic equation of state,

$$\pi_T = T \left(\frac{\partial p}{\partial T}\right)_V - p = T \left(\frac{nR}{V - nb}\right) - p = \frac{nRT}{V - nb} - \left(\frac{nRT}{V - nb} - a \frac{n^2}{V^2}\right)$$

$$\pi_T = a \frac{n^2}{V^2}$$

Thermochemistry of G

$$G = H - TS$$

At constant T ,

$$\Delta G = \Delta H - T\Delta S$$

We can define the **standard Gibbs energy of reaction**:

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

and the **standard Gibbs energy of formation**:

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

Standard Reaction Potentials

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

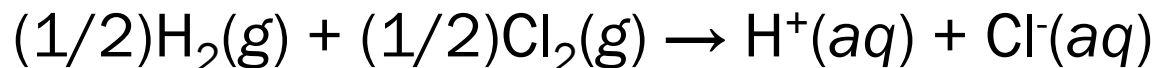
$$\Delta_r S^\ominus = \sum_{\text{products } i} \nu(i) S_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) S_m^\ominus(j)$$

$$\Delta_r G^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f G^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f G^\ominus(j)$$

Convention Again

$$\Delta_f G^\ominus(\text{H}^+, \text{aq}) = 0$$

e.g. (Brief Illustration 3D.2)



$$\Delta_r G^\ominus = \Delta_f G^\ominus(\text{H}^+, \text{aq}) + \Delta_f G^\ominus(\text{Cl}^-, \text{aq}) = \Delta_f G^\ominus(\text{Cl}^-, \text{aq})$$

Pressure Dependence of G

$$dG = Vdp - SdT$$

At constant T ,

$$dG = Vdp$$

Integration leads to

$$\int_{\text{state } i}^{\text{state } f} dG = \int_{\text{state } i}^{\text{state } f} V dp$$

$$\Delta G = G_f - G_i = \int_{p_i}^{p_f} V dp$$

Pressure Dependence of G

$$\Delta G = \int_{p_i}^{p_f} V dp$$

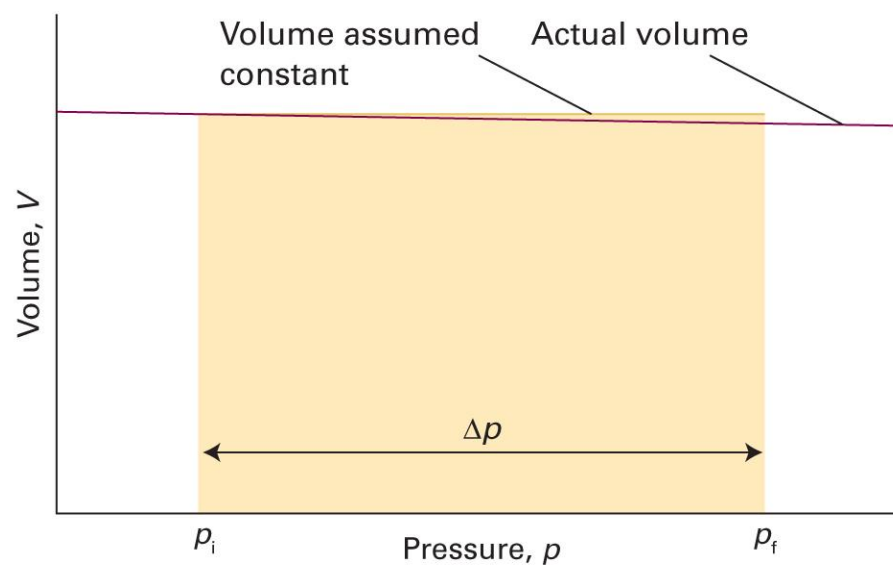
Dividing both sides by n ,

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp$$

1. Condensed phase: $V_m \approx \text{constant}$
2. Gas phase: $V_m = V/n = RT/p$

Condensed Phase

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp = V_m \int_{p_i}^{p_f} dp = V_m(p_f - p_i)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 3E.4)

Example 3E.3

Suppose that for a certain phase transition of a solid $\Delta_{\text{trs}}V_m = +1.0 \text{ cm}^3 \text{ mol}^{-1}$ independent of pressure. By how much does that Gibbs energy of transition change when the pressure is increased from 1.0 bar ($1.0 \times 10^5 \text{ Pa}$) to 3.0 Mbar ($3.0 \times 10^{11} \text{ Pa}$)?

For the solid phase,

$$\Delta G_{m,s} = G_{m,s}(p_f) - G_{m,s}(p_i) = V_{m,s}(p_f - p_i)$$

For the liquid phase,

$$\Delta G_{m,l} = G_{m,l}(p_f) - G_{m,l}(p_i) = V_{m,l}(p_f - p_i)$$

Subtraction gives

$$\begin{aligned} \Delta_{\text{trs}}G_m(p_f) - \Delta_{\text{trs}}G_m(p_i) &= (V_{m,s} - V_{m,l})(p_f - p_i) \\ &= \Delta_{\text{trs}}V_m \times (p_f - p_i) \end{aligned}$$

Example 3E.3

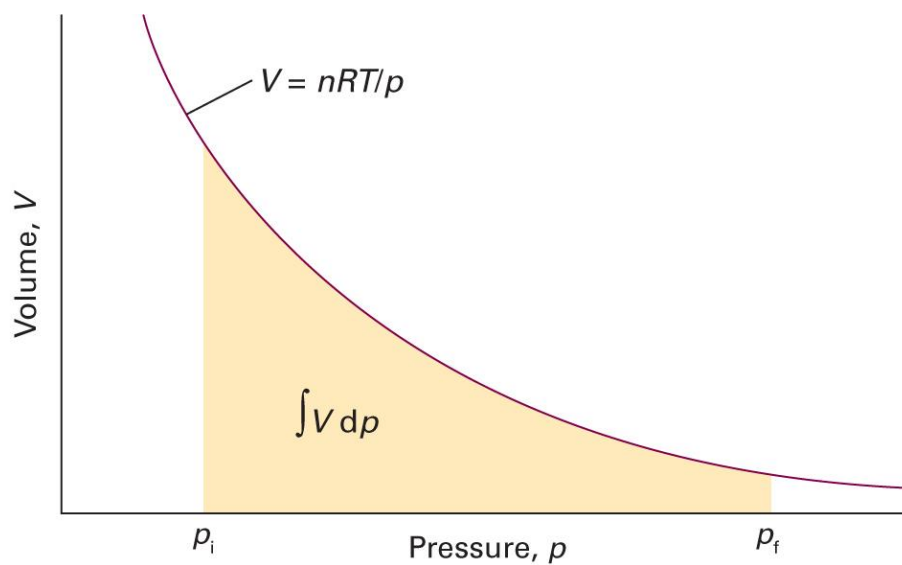
$$\Delta_{\text{trs}}G_m(p_f) - \Delta_{\text{trs}}G_m(p_i) = \Delta_{\text{trs}}V_m \times (p_f - p_i)$$

Hence,

$$\begin{aligned} & \Delta_{\text{trs}}G_m(3.0 \text{ Mbar}) - \Delta_{\text{trs}}G_m(1.0 \text{ bar}) \\ &= (1.0 \text{ cm}^3 \text{ mol}^{-1}) \times (3.0 \times 10^{11} \text{ Pa} - 1.0 \times 10^5 \text{ Pa}) \\ &= 3.0 \times 10^2 \text{ kJ mol}^{-1} \end{aligned}$$

Gas Phase

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp = \int_{p_i}^{p_f} \frac{RT}{p} dp = RT(\ln p_f - \ln p_i)$$



Brief Illustration 3E.2

When the pressure is increased isothermally on water vapour (treated as a perfect gas) from 1.0 bar to 2.0 bar at 298 K, what is the change in molar Gibbs energy?

$$\Delta G_m = RT(\ln p_f - \ln p_i) = RT \ln \frac{p_f}{p_i}$$

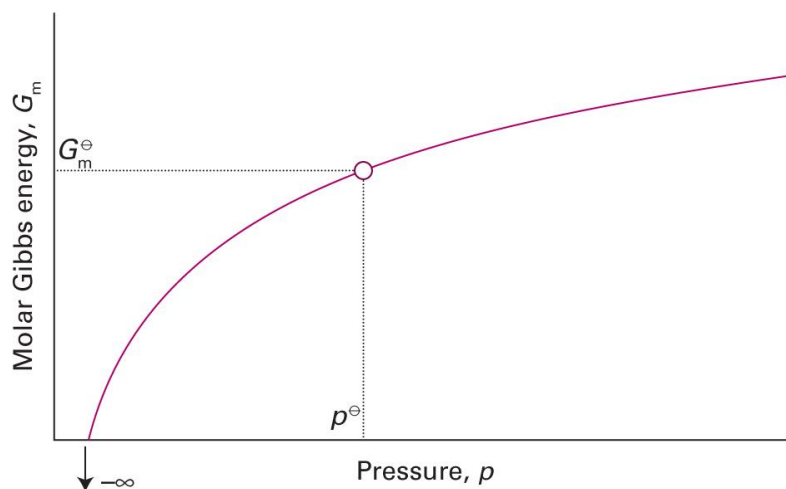
Substitution leads to

$$\begin{aligned}\Delta G_m &= (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times \ln (2.0 \text{ bar} / 1.0 \text{ bar}) \\ &= 1.72 \text{ kJ mol}^{-1}\end{aligned}$$

Gas Phase: Standard State

$$\Delta G_m = G_m(p_f) - G_m(p_i) = RT \ln \frac{p_f}{p_i}$$

$$G_m(p) = G_m^\ominus + RT \ln \frac{p}{p^\ominus}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 3E.6)

Summary

Thermodynamic potentials

Clausius ineq.

First law

Internal energy

$$U$$

Helmholtz energy

$$A = U - TS$$

Enthalpy

$$H = U + pV$$

Gibbs energy

$$G = H - TS$$

Spontaneity

$$\Delta A_{T,V} \leq 0$$

$$\Delta G_{T,p} \leq 0$$

Maximum work

$$A_T = w_{\max}$$

$$G_{T,p} = w_{\text{add,max}}$$

Differential forms

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

